

```
from google.colab import files
uploaded = files.upload()
```

Choose Files 2 files

True.csv(text/csv) - 53582940 bytes, last modified: 4/24/2026 - 100% done
Fake.csv(text/csv) - 62789876 bytes, last modified: 4/24/2026 - 100% done
 Saving True.csv to True.csv
 Saving Fake.csv to Fake.csv

```
import pandas as pd

fake_df = pd.read_csv("Fake.csv")
real_df = pd.read_csv("True.csv")

print(fake_df.head())
print(real_df.head())
```

```

                                title \
0  Donald Trump Sends Out Embarrassing New Year'...
1  Drunk Bragging Trump Staffer Started Russian ...
2  Sheriff David Clarke Becomes An Internet Joke...
3  Trump Is So Obsessed He Even Has Obama's Name...
4  Pope Francis Just Called Out Donald Trump Dur...

                                text subject \
0  Donald Trump just couldn t wish all Americans ...   News
1  House Intelligence Committee Chairman Devin Nu...   News
2  On Friday, it was revealed that former Milwauk...   News
3  On Christmas day, Donald Trump announced that ...   News
4  Pope Francis used his annual Christmas Day mes...   News

                                date
0  December 31, 2017
1  December 31, 2017
2  December 30, 2017
3  December 29, 2017
4  December 25, 2017

                                title \
0  As U.S. budget fight looms, Republicans flip t...
1  U.S. military to accept transgender recruits o...
2  Senior U.S. Republican senator: 'Let Mr. Muell...
3  FBI Russia probe helped by Australian diplomat...
4  Trump wants Postal Service to charge 'much mor...

                                text          subject \
0  WASHINGTON (Reuters) - The head of a conservat...  politicsNews
1  WASHINGTON (Reuters) - Transgender people will...  politicsNews
2  WASHINGTON (Reuters) - The special counsel inv...  politicsNews
3  WASHINGTON (Reuters) - Trump campaign adviser ...  politicsNews
4  SEATTLE/WASHINGTON (Reuters) - President Donal...  politicsNews

                                date
0  December 31, 2017
1  December 29, 2017
2  December 31, 2017
3  December 30, 2017
4  December 29, 2017
```

```
fake_df["label"] = 1
real_df["label"] = 0
```

```
df = pd.concat([fake_df, real_df], axis=0)
df = df.sample(frac=1, random_state=42).reset_index(drop=True) # shuffle

print(df.shape)
df.head()
```

(44898, 5)

	title	text	subject	date	label	
0	Ben Stein Calls Out 9th Circuit Court: Commit...	21st Century Wire says Ben Stein, reputable pr...	US_News	February 13, 2017	1	
1	Trump drops Steve Bannon from National Securit...	WASHINGTON (Reuters) - U.S. President Donald T...	politicsNews	April 5, 2017	0	
2	Puerto Rico expects U.S. to lift Jones Act shi...	(Reuters) - Puerto Rico Governor Ricardo Rosse...	politicsNews	September 27, 2017	0	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df = df[["title", "text", "label"]]
df["content"] = df["title"].fillna("") + " " + df["text"].fillna("")
```

```
import re

def clean_text(s):
    s = s.lower()
    s = re.sub(r"http\S+|www\S+", "", s)
    s = re.sub(r"^\a-z\s]", " ", s)
    s = re.sub(r"\s+", " ", s).strip()
    return s

df["clean_content"] = df["content"].apply(clean_text)
```

```
from sklearn.model_selection import train_test_split

X = df["clean_content"]
y = df["label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import classification_report

pipe = Pipeline([
    ("tfidf", TfidfVectorizer(max_features=5000, ngram_range=(1,2))),
    ("clf", LogisticRegression(max_iter=1000))
])

pipe.fit(X_train, y_train)
y_pred = pipe.predict(X_test)

print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	4284
1	0.99	0.99	0.99	4696
accuracy			0.99	8980
macro avg	0.99	0.99	0.99	8980
weighted avg	0.99	0.99	0.99	8980

```
!pip install lime
from lime.lime_text import LimeTextExplainer

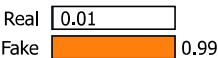
class_names = ["Real", "Fake"]
explainer = LimeTextExplainer(class_names=class_names)

def predict_proba(texts):
    return pipe.predict_proba(texts)

exp = explainer.explain_instance(X_test.iloc[0], predict_proba, num_features=10)
exp.show_in_notebook(text=True)
```

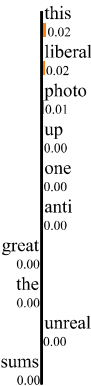
```
Collecting lime
  Downloading lime-0.2.0.1.tar.gz (275 kB)
    275.7/275.7 kB 5.8 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from lime) (3.10.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from lime) (2.0.2)
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Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/python3.12/dist-packages (from scikit-image>=0.12->lime) (2022.8.12)
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Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image>=0.12->lime) (0.5)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->lime) (1.17.0)
Building wheels for collected packages: lime
  Building wheel for lime (setup.py) ... done
  Created wheel for lime: filename=lime-0.2.0.1-py3-none-any.whl size=283834 sha256=119c77ef166a10cdee2e8ea16479c616f4d68496ea15
  Stored in directory: /root/.cache/pip/wheels/e7/5d/0e/4b4fff9a47468fed5633211fb3b76d1db43fe806a17fb7486a
Successfully built lime
Installing collected packages: lime
Successfully installed lime-0.2.0.1
```

Prediction probabilities



Real

Fake



Text with highlighted words

unreal this one photo sums up the anti trump liberal
wackos great parenting huh

```
real_news = ""
The Indian government has announced a new initiative to expand renewable energy capacity
across rural regions. The policy focuses on increasing solar power installations and
improving grid infrastructure. Experts believe this step will help reduce carbon emissions
and promote sustainable development in the coming years.
""
```

```
# Predict using your trained pipeline
pred = pipe.predict([real_news])[0]
prob = pipe.predict_proba([real_news])[0]

print("Prediction:", "Fake" if pred==1 else "Real")
print("Confidence (Real, Fake):", prob)
```

Prediction: Fake
Confidence (Real, Fake): [0.37088191 0.62911809]

```
!pip install lime

from lime.lime_text import LimeTextExplainer

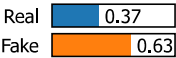
explainer = LimeTextExplainer(class_names=["Real", "Fake"])

exp = explainer.explain_instance(real_news, pipe.predict_proba, num_features=10)

# Show explanation
exp.show_in_notebook(text=True)
```

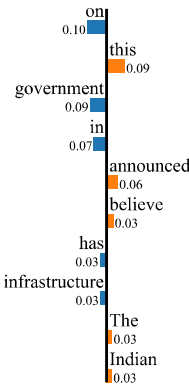
Requirement already satisfied: lime in /usr/local/lib/python3.12/dist-packages (0.2.0.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from lime) (3.10.0)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->lime) (1.16.0)

Prediction probabilities



Real

Fake



Text with highlighted words

The Indian government has announced a new initiative to expand renewable energy capacity across rural regions. The policy focuses on increasing solar power installations and improving grid infrastructure. Experts believe this step will help reduce carbon emissions and promote sustainable development in the coming years.

```
news_1 = ""
The Reserve Bank of India has decided to maintain the current interest rates in order
to control inflation and ensure economic stability. Officials stated that the decision
was based on recent economic data and global financial trends.
""
```

```
news_2 = ""
Experts have raised concerns over the increasing levels of air pollution in major cities.
They warned that if immediate action is not taken, public health could be severely affected
in the coming years, especially among children and elderly populations.
""
```

```
news_3 = ""
Shocking reports reveal that the government is hiding dangerous secrets about pollution levels.
Experts claim this could destroy entire cities and put millions of lives at risk overnight.
This unbelievable situation has left citizens terrified and demanding immediate answers.
""
```

```
from lime.lime_text import LimeTextExplainer

explainer = LimeTextExplainer(class_names=["Real", "Fake"])

news_samples = {
    "News 1 (Neutral)": news_1,
    "News 2 (Slightly Emotional)": news_2,
    "News 3 (Exaggerated)": news_3
}

for name, text in news_samples.items():
    print("\n=====")
    print(name)
    print("=====")

    # Prediction
    pred = pipe.predict([text])[0]
    prob = pipe.predict_proba([text])[0]
```

```
print("Prediction:", "Fake" if pred==1 else "Real")
print("Confidence (Real, Fake):", prob)

# Explanation
exp = explainer.explain_instance(text, pipe.predict_proba, num_features=8)
exp.show_in_notebook(text=True)
```

```
=====
News 1 (Neutral)
=====
```

Prediction: Fake

Confidence (Real, Fake): [0.33361082 0.66638918]

Prediction probabilities



Real

Fake



Text with highlighted words

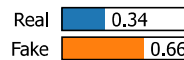
The Reserve Bank of India has decided to maintain the current interest rates in order to control inflation and ensure economic stability. Officials stated that the decision was based on recent economic data and global financial trends.

```
=====
News 2 (Slightly Emotional)
=====
```

Prediction: Fake

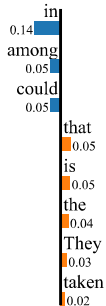
Confidence (Real, Fake): [0.34342649 0.65657351]

Prediction probabilities



Real

Fake



Text with highlighted words

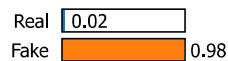
Experts have raised concerns over the increasing levels of air pollution in major cities. They warned that if immediate action is not taken, public health could be severely affected in the coming years, especially among children and elderly populations.

```
=====
News 3 (Exaggerated)
=====
```

Prediction: Fake

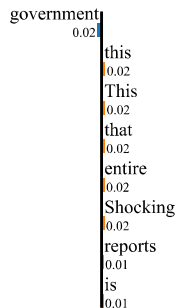
Confidence (Real, Fake): [0.0215088 0.9784912]

Prediction probabilities



Real

Fake



Text with highlighted words

Shocking reports reveal that the government is hiding dangerous secrets about pollution levels. Experts claim this could destroy entire cities and put millions of lives at risk overnight. This unbelievable situation has left citizens terrified and demanding immediate answers.

```
test_news_real = ""
```

```
The government said on Monday it will continue to support economic growth through
policy measures aimed at improving infrastructure and boosting investment. Officials
stated that the decision was based on recent economic indicators and global market trends.
""
```

```
pred = pipe.predict([test_news_real])[0]
prob = pipe.predict_proba([test_news_real])[0]
```

```
print("Prediction:", "Fake" if pred==1 else "Real")
print("Confidence (Real, Fake):", prob)
```

```
Prediction: Real
Confidence (Real, Fake): [0.92713204 0.07286796]
```

```
!pip install lime

from lime.lime_text import LimeTextExplainer

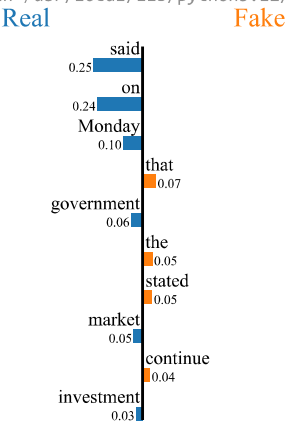
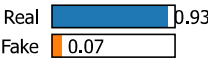
explainer = LimeTextExplainer(class_names=["Real", "Fake"])

exp = explainer.explain_instance(test_news_real, pipe.predict_proba, num_features=10)

exp.show_in_notebook(text=True)
```

Requirement already satisfied: lime in /usr/local/lib/python3.12/dist-packages (0.2.0.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from lime) (3.10.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from lime) (2.0.2)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->lime) (1.17.0)

Prediction probabilities



Text with highlighted words

The government said on Monday it will continue to support economic growth through policy measures aimed at improving infrastructure and boosting investment. Officials stated that the decision was based on recent economic indicators and global market trends.

Input Type	Style	Prediction	Observation
Real news (different style)	Neutral	Fake ❌	Misclassified
Real news (Reuters style)	Formal	Real ✅	Correct
Exaggerated news	Emotional	Fake ✅	Correct

```
Fake_news = """information that is verified, credible, and published by trusted sources, while fake news is false or misleading
```

```
pred = pipe.predict([Fake_news])[0]
prob = pipe.predict_proba([Fake_news])[0]

print("Prediction:", "Fake" if pred==1 else "Real")
print("Confidence (Real, Fake):", prob)
```

Prediction: Fake
Confidence (Real, Fake): [0.19642643 0.80357357]

```
news_4 = """
The Ministry of Health announced on Tuesday that it will expand vaccination programs
in rural areas to improve healthcare access. Officials said the initiative aims to
increase immunization coverage and reduce the spread of preventable diseases across
the country.
"""
```

```
pred = pipe.predict([news_4])[0]
prob = pipe.predict_proba([news_4])[0]

print("Prediction:", "Fake" if pred==1 else "Real")
print("Confidence (Real, Fake):", prob)
```

Prediction: Real
Confidence (Real, Fake): [0.92970682 0.07029318]

```
!pip install lime

from lime.lime_text import LimeTextExplainer

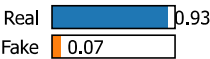
explainer = LimeTextExplainer(class_names=["Real", "Fake"])

exp = explainer.explain_instance(news_4, pipe.predict_proba, num_features=10)

exp.show_in_notebook(text=True)
```

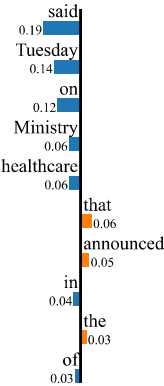
Requirement already satisfied: lime in /usr/local/lib/python3.12/dist-packages (0.2.0.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from lime) (3.10.0)
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Requirement already satisfied: pillow>=10.1 in /usr/local/lib/python3.12/dist-packages (from scikit-image>=0.12->lime) (11.3.0)
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Prediction probabilities



Real

Fake



Text with highlighted words

The Ministry of Health announced on Tuesday that it will expand vaccination programs in rural areas to improve healthcare access. Officials said the initiative aims to increase immunization coverage and reduce the spread of preventable diseases across the country.

```
news_large = """
The government on Monday announced a series of measures aimed at strengthening economic growth across multiple sectors.
Officials stated that the policy framework focuses on infrastructure development, job creation, and sustainable investment stra
The finance ministry highlighted that increased public spending would be directed toward transportation, energy, and digital in
Experts believe that these initiatives could improve long-term economic stability and attract foreign investment.
In addition, the policy includes incentives for small and medium enterprises to expand their operations.
Industry leaders have welcomed the decision, noting that it could boost confidence in the domestic market.
The central bank also indicated that it would continue to monitor inflation and adjust interest rates accordingly.
"""
```


Analysts observed that maintaining a balance between growth and inflation control remains a key challenge. Meanwhile, the government emphasized the importance of innovation and technology adoption in driving economic progress. Programs supporting startups and research institutions are expected to receive additional funding. Officials confirmed that collaboration between public and private sectors will play a critical role in implementation. The policy also outlines measures to improve ease of doing business and reduce regulatory barriers. According to reports, several reforms are already in progress to streamline approval processes. Economic experts noted that consistent policy execution will be essential for achieving desired outcomes. The government reiterated its commitment to fiscal discipline while pursuing development goals. International observers have responded positively to the announcement, citing strong economic fundamentals. Market reactions remained stable following the release of the policy details. Financial analysts suggested that long-term benefits would depend on effective execution. The initiative also aims to address regional disparities by focusing on rural and underdeveloped areas. Infrastructure projects in these regions are expected to generate employment opportunities. Officials stated that improving connectivity will support trade and local industries. The plan includes modernization of existing transport networks and expansion of new routes. Energy sector reforms are also part of the broader strategy. Renewable energy projects are expected to receive increased attention and investment. Experts highlighted the importance of sustainability in future economic planning. Environmental considerations have been integrated into policy decisions. The government emphasized that growth and sustainability must go hand in hand. Healthcare and education sectors are also expected to benefit from increased funding. Officials noted that human capital development is essential for long-term progress. Efforts are being made to improve access to quality education and training programs. Healthcare infrastructure improvements are aimed at better service delivery. The policy also includes provisions for digital transformation across sectors. Expansion of internet connectivity and digital services is a key focus area. Experts believe that digital inclusion will enhance productivity and innovation. The government stated that transparency and accountability will be ensured throughout implementation. Regular monitoring and evaluation mechanisms have been established. Stakeholders will be engaged to provide feedback and suggestions. Public awareness campaigns are planned to inform citizens about the new initiatives. Officials expressed confidence in achieving the objectives set under the policy. Economic indicators will be closely tracked to assess progress. The government assured that corrective measures would be taken if required. Overall, the policy represents a comprehensive approach to economic development. Experts concluded that sustained efforts and coordinated action will be necessary for success.

"""

```
pred = pipe.predict([news_large])[0]
prob = pipe.predict_proba([news_large])[0]

print("Prediction:", "Fake" if pred==1 else "Real")
print("Confidence:", prob)
```

```
Prediction: Real
Confidence: [0.84143899 0.15856101]
```

```
!pip install lime

from lime.lime_text import LimeTextExplainer

explainer = LimeTextExplainer(class_names=["Real", "Fake"])

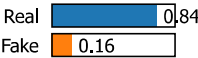
exp = explainer.explain_instance(news_large, pipe.predict_proba, num_features=10)

exp.show_in_notebook(text=True)
```



```
Requirement already satisfied: lime in /usr/local/lib/python3.12/dist-packages (0.2.0.1)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->lime)
```

Prediction probabilities



Real

Fake



Text with highlighted words

The government on Monday announced a series of